

# RAHIM HOSSAIN

Applied Machine Learning Researcher  
Multimodal Retrieval and Vision Language Models

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## SUMMARY

Applied machine learning researcher and Computer Science PhD candidate specializing in multimodal retrieval and vision-language models. Industry experience in large-scale product search, model adaptation, and evaluation at eBay. Publications at ICCV, IJCNN, and Sensors; first-author manuscripts under review at NeurIPS 2026 and in submission to ICLR 2027.

## EXPERIENCE

### eBay

San Jose, CA | May 2026 - Aug 2026

*Applied Research Intern, Ads Team*

- Fine-tuned BLIP with visual-grounding late interaction on eBay Fashion catalog data, improving retrieval over the existing production model.
- Built end-to-end data and evaluation pipelines and ran large-scale training on labeled product data using eBay's GPU clusters.
- Designed Tip-Region, a gradient-free few-shot adaptation method that augments frozen backbones with a paired text-region exemplar cache. Improved localized retrieval at every evaluated recall level across CLIP, SigLIP2, and Qwen3-VL-Embedding.
- Identified and corrected train-test contamination in a public benchmark mirror through a support-set leakage audit; built LocFG, a leakage-controlled benchmark for small-referent retrieval.

### Binghamton University (SUNY)

Binghamton, NY | May 2023 - Present

*Graduate Research Assistant, RT Lab*

- Developed MM++, an unsupervised, scale-invariant method for multilayer out-of-distribution (OOD) detection using top-K gated feature fusion.
- Built an energy-efficient deep learning pipeline for spotted-lanternfly detection over LoRa wide-area networks.

### Cobalt Speech & Language, Inc.

Remote | Mar 2022 - Dec 2022

*Research Scientist*

- Fine-tuned and evaluated BART, BERT, and LED transformer models for long-context summarization and natural-language understanding as a core contributor to the Call Summarization product.

## EDUCATION

### Binghamton University (SUNY)

Aug 2024 - Present

PhD in Computer Science

### Binghamton University (SUNY)

Jan 2023 - May 2024

MS in Computer Science

### Bangladesh University of Engineering & Technology (BUET)

Apr 2016 - Jan 2022

BSc in Electrical and Electronic Engineering

## SELECTED PUBLICATIONS AND MANUSCRIPTS

### Manuscripts

**Tip-Region.** First-author manuscript on gradient-free few-shot adaptation for localized vision-language retrieval, based on eBay internship research. *In submission to ICLR 2027.*

**R. Hossain** et al. "MM++: Unsupervised Scale-Invariant Multilayer OOD Detection via Top-K Gated Feature Fusion." *Under review at NeurIPS 2026.* [arXiv:2606.17352](https://arxiv.org/abs/2606.17352).

### Published Papers

S. Ahmed, A. A. Arafat, M. N. Rizve, **R. Hossain**, Z. Guo, A. S. Rakin. "SSDA: Secure Source-Free Domain Adaptation." *ICCV*, 2023.

M. D. Samad, **R. Hossain**, K. M. Iftekharuddin. "Dynamic Perturbation of Weights for Improved Data Reconstruction in Unsupervised Learning." *IJCNN*, 2021.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Go, MATLAB

**Frameworks:** PyTorch, Hugging Face Transformers, TensorFlow, Keras, scikit-learn

**Research Areas:** Multimodal retrieval; vision-language models (CLIP, SigLIP/SigLIP2, BLIP, Qwen-VL); late interaction; few-shot and test-time adaptation; NLP and LLMs

**Infrastructure:** Distributed training on GPU clusters; large-scale data and evaluation pipelines; Linux

## AWARDS AND HONORS

- **eBay Vibe Coding Hackathon, Champion (2026).** Won among all intern teams company-wide; recognized by CEO Jamie Iannone, with follow-up engagement from eBay's marketing team.
- **eBay ML Challenge.** 2nd place (2025); 3rd place (2024).
- **Industrial Automation Challenge.** 2nd place in a national robotics competition in Bangladesh.

## REFERENCES

### Kyoung-Don Kang

Professor, School of Computing, Binghamton University (SUNY)

PhD advisor | [kang@binghamton.edu](mailto:kang@binghamton.edu)

### Xinxin Shu

Manager, eBay Ads, eBay Inc.

Internship supervisor | [xshu@ebay.com](mailto:xshu@ebay.com)

### Adnan Siraj Rakin

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Research collaborator and supervisor | [arakin@binghamton.edu](mailto:arakin@binghamton.edu)